Explainable AI for Network Intrusion Detection Systems: Exploring Performance–Explainability Trade-offs in Ensemble Learning

Himaja
Student ID: 24279048
H9AIMLC – AI/ML in Cybersecurity
National College of Ireland
Dublin, Ireland

***Abstract*—Network Intrusion Detection Systems (NIDS) based on machine learning ensemble techniques achieve 97–99% detection accuracy but remain largely undeployed in Security Operations Centres (SOCs) due to their black-box nature. Regulatory frameworks including GDPR Article 22 and the EU AI Act mandate explanations for automated decisions, making explainability a legal requirement. This study systematically evaluates the trade-off between detection performance and explanation quality across three ensemble architectures—bagging (Random Forest), boosting (XGBoost, LightGBM), and stacking—combined with two post-hoc explainability methods (SHAP and LIME) on the UNSW-NB15 benchmark dataset. Three novel metrics are introduced: a Consistency Index measuring SHAP–LIME agreement via Kendall's τ, Explanation Fidelity quantifying correlation between feature importance and accuracy degradation, and Computational Overhead Ratio capturing explanation latency relative to prediction time. Results demonstrate that boosting architectures provide the best balance of detection accuracy and explainability, with SHAP offering superior computational efficiency over LIME for tree-based ensembles.**

***Index Terms*—Explainable AI, Network Intrusion Detection, Ensemble Learning, SHAP, LIME, UNSW-NB15**

I. Introduction

Network Intrusion Detection Systems (NIDS) based on machine learning ensemble techniques have demonstrated 97–99% accuracy in detecting network intrusions. However, due to their black-box nature, these methods are not yet widely adopted within Security Operations Centres (SOCs). Research indicates that 40–60% of alerts generated by ML-based NIDS are re-checked by analysts because they cannot ascertain how or why a decision was reached [1]. Regulatory compliance further compounds this issue; both GDPR Article 22 and the EU AI Act (2024) require explanations for automated decisions, rendering explainability a legal requirement rather than merely a desirable characteristic [2].

Post-hoc explainability methods such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) provide meaningful insights into feature utilisation. However, their application within NIDS creates computational burdens and raises concerns about explanation consistency, fidelity, and operational applicability [3].

The existing literature reveals a significant evidence gap: no comprehensive examination of the relationship between ensemble architectures (bagging, boosting, stacking) and XAI methods has been conducted. Each architecture type differently affects model complexity and the manner in which explanations are generated. Understanding this intersection is critical for providing evidence-based recommendations for both operational deployment and regulatory compliance.

This study addresses the research question: *How do Bagging, Boosting, and Stacking ensemble methods, when used with SHAP and LIME, affect the trade-off between detection performance (accuracy, F1-score) and explanation quality (consistency, fidelity, latency) in network intrusion detection systems, and what are the implications for GDPR Article 22 and EU AI Act compliance?*

The UNSW-NB15 benchmark dataset is employed, containing over 2,000,000 network flows across 43 features and 9 attack categories, providing a comprehensive evaluation basis.

II. Related Work

A. Foundations of Post-Hoc Explainability

Lundberg and Lee [4] introduced SHAP (SHapley Additive exPlanations), unifying six existing feature-attribution methods under a single additive framework grounded in cooperative game theory. Their key contribution is demonstrating that Shapley values [5] are the unique solution satisfying the axioms of local accuracy, missingness, and consistency—properties that no other additive attribution method jointly guarantees. However, the computational cost of exact Shapley value computation is exponential in the number of features, which motivates the TreeExplainer approximation used in the present study. Ribeiro et al. [6] proposed LIME (Local Interpretable Model-agnostic Explanations), which constructs a locally faithful linear surrogate by perturbing inputs around a query instance and fitting a weighted regression. LIME's model-agnostic design makes it applicable to any classifier, but its reliance on random perturbations introduces stochasticity that can reduce explanation stability across runs. The theoretical guarantee underpinning SHAP traces back to Shapley's original formulation [5], which establishes that the Shapley

value is the only allocation satisfying efficiency, symmetry, dummy player, and additivity axioms—providing a stronger axiomatic foundation than LIME's empirical fidelity objective.

B. NIDS Benchmarks and Classical Approaches

The UNSW-NB15 dataset employed in this study was introduced by Moustafa and Slay [7], who designed it to address the well-documented limitations of older benchmarks such as KDD Cup 99. Moustafa and Slay generated realistic modern attack traffic using the IXIA PerfectStorm tool across nine attack categories, producing over two million records with 49 features. Earlier, Tavallaee et al. [8] conducted a detailed statistical analysis of the KDD Cup 99 dataset (producing the NSL-KDD variant), revealing that approximately 78% of training records and 75% of test records were duplicates, which inflated classifier accuracy and biased evaluations toward majority classes. The transition from NSL-KDD to UNSW-NB15 represents a maturation of the NIDS evaluation landscape, though both datasets remain synthetic benchmarks whose class separability may not reflect the difficulty of production network traffic. Ferrag et al. [9] surveyed deep learning approaches across seven NIDS benchmark datasets and found that ensemble and gradient-boosted tree methods consistently rivalled or outperformed deep architectures on tabular network-flow data, reinforcing the relevance of the ensemble paradigms examined in the present study.

C. Explanation Consistency and Agreement

Krishna et al. [10] empirically demonstrated that different post-hoc explanation methods frequently disagree on feature importance rankings for the same model and instance—a phenomenon they term the "disagreement problem." Their study found that SHAP, LIME, and gradient-based methods produced substantially different top-k feature sets in up to 40% of test instances across multiple domains. This finding directly motivates the Consistency Index introduced in the present study: if practitioners cannot assume that explanation methods agree, then quantifying disagreement via Kendall's τ becomes essential for responsible deployment. Krishna et al. further showed that disagreement is more pronounced for complex models with many interacting features, suggesting that ensemble NIDS—which inherently exploit feature interactions—are particularly susceptible.

D. Regulatory Requirements for Explainability

The General Data Protection Regulation [11] establishes in Article 22 that data subjects have the right not to be subject to decisions based solely on automated processing unless accompanied by meaningful information about the logic involved. The EU Artificial Intelligence Act [12] extends this requirement by classifying high-risk AI systems—a category that may encompass automated security monitoring—and mandating that such systems provide sufficient transparency for users to interpret and appropriately use their outputs. Together, these regulations create a legal obligation for NIDS operators to produce human-readable explanations, elevating XAI from a research convenience to a compliance necessity. The present study operationalises this obligation by measuring whether SHAP and LIME can deliver consistent, timely explanations suitable for regulatory audit.

E. Adversarial Risks to Explanations

Slack et al. [13] demonstrated that post-hoc explanation methods including LIME and SHAP can be deliberately manipulated by adversarial classifiers that detect when an input is a perturbation query and alter their behaviour to produce misleading feature attributions. Their scaffolded classifier attack successfully hid biased decision-making from both LIME and SHAP audits, raising serious concerns about the trustworthiness of explanations in adversarial settings. In the NIDS domain, this vulnerability is particularly acute because sophisticated attackers could craft models or traffic patterns designed to defeat not only the detection system but also its explanation pipeline, thereby evading both automated and human-in-the-loop oversight.

F. XAI-Integrated NIDS Studies

Alabbadi and Bajaber [14] demonstrated effective use of integrated LIME with a Random Forest IoT intrusion detection system, reducing false positive rates by 20% and improving analyst triage decision-making. However, the study focuses solely on IoT network traffic using a single model without SHAP comparisons or quantitative assessment of performance-to-explainability relationships.

Alamro et al. [15] achieved 99.1% overall accuracy through deep learning with dimensionality reduction but omitted XAI integration, resulting in opaque latent representations that violate both GDPR Article 22 and EU AI Act requirements.

Fatema et al. [16] implemented federated learning within an integrated SHAP system achieving 98.50% accuracy while maintaining data decentralisation. However, the federated structure creates obstacles to ensemble comparison, and SHAP was used without systematic evaluation against alternative explanation methods.

Zibi and Obagbuwa [17] achieved 97.8% accuracy with SHAP on XGBoost but did not conduct cross-validation with LIME or ablation studies. Separately, their systematic review [3] supported XAI as essential for analyst trust but did not provide objective measures to define explanation quality.

Khan et al. [2] comprehensively surveyed the XAI-IDS landscape, identifying the absence of multi-architecture evaluation and adversarial risks for XAI, but the survey provides no experimental validation of identified gaps.

Samed and Sagiroglu [18] reviewed SHAP, LIME, and Grad-CAM, finding SHAP to be the most widely used method, but the review treats algorithms as interchangeable without empirical trade-off analysis.

Pawlicki et al. [1] identified latency constraints, false-positive costs, and adversarial risk as the three major challenges of XAI-IDS integration, but all findings remained theoretical without architectural solutions.

G. Summary of Gaps

Collectively, the literature reveals four unaddressed gaps that the present study targets: (1) no prior work has systematically compared bagging, boosting, and stacking ensembles under a unified XAI evaluation framework; (2) explanation consistency across methods remains unquantified in the NIDS domain, despite the disagreement problem documented by Krishna et al. [10]; (3) computational overhead of explanation generation—critical for SOC deployment—has not been benchmarked across ensemble families; and (4) regulatory compliance implications of XAI method selection remain discussed only qualitatively, without empirical metrics.

III. Methodology and Implementation

A. Dataset and Preprocessing

The UNSW-NB15 dataset [7] is preprocessed through: (i) KNN imputation for missing values; (ii) Winsorisation for outlier treatment; (iii) target encoding for categorical features; (iv) Robust Scaler normalisation; (v) SMOTE for class imbalance correction; and (vi) VIF-based multicollinearity removal. The dataset is split into 60% training, 20% validation, and 20% test sets with stratified sampling.

B. Statistical Validation Protocol

To ensure that reported metrics are robust and not artefacts of a single data split, all experiments are repeated across three independent random seeds (42, 123, 456) using 5-fold stratified cross-validation on the training partition. The test-set results reported in Table I represent the mean across seeds; the observed standard deviations for accuracy and F1-score are ± 0.000 for all four architectures, confirming that variance is negligible on this benchmark. For the Consistency Index (τ), inter-seed standard deviations are ± 0.008 (Random Forest), ± 0.006 (XGBoost), ± 0.011 (LightGBM), and ± 0.004 (Stacking), indicating that the rank-order differences between architectures are stable and not attributable to sampling noise. We do not report formal statistical significance tests (e.g., paired t-tests) for classification metrics because the ceiling effect renders all architectures indistinguishable on this dataset; however, for the Consistency Index, the observed τ differences between Random Forest (0.171 ± 0.008) and LightGBM (-0.082 ± 0.011) exceed three times the pooled standard deviation, providing strong evidence that the architectural effect on SHAP–LIME agreement is genuine rather than stochastic.

C. Ensemble Architectures

Three ensemble paradigms are evaluated:

Bagging (Random Forest): 200 estimators with max depth 20, using balanced class weights.

Boosting: XGBoost (200 estimators, max depth 8, learning rate 0.1) and LightGBM (200 estimators, max depth 8, learning rate 0.1) represent gradient boosting approaches.

Stacking: A meta-learner combining Random Forest, XGBoost, and LightGBM base learners with Logistic Regression as the final estimator, trained with 3-fold cross-validation.

D. Explainability Methods

SHAP: TreeExplainer is used for individual tree-based models, and KernelExplainer for the stacking ensemble. Mean absolute SHAP values provide global feature importance.

LIME: LimeTabularExplainer generates local explanations for individual predictions, aggregated across test samples to derive global importance rankings.

E. Evaluation Metrics

Three novel XAI quality metrics are defined:

Consistency Index: Kendall's τ rank correlation between SHAP and LIME feature importance rankings, measuring agreement between explanation methods.

Explanation Fidelity: Correlation between feature importance rankings and accuracy degradation upon sequential feature removal, measuring whether important features genuinely drive model decisions.

Computational Overhead Ratio: The ratio of explanation generation time to prediction time, quantifying the operational cost of explainability.

IV. Evaluation

A. Results

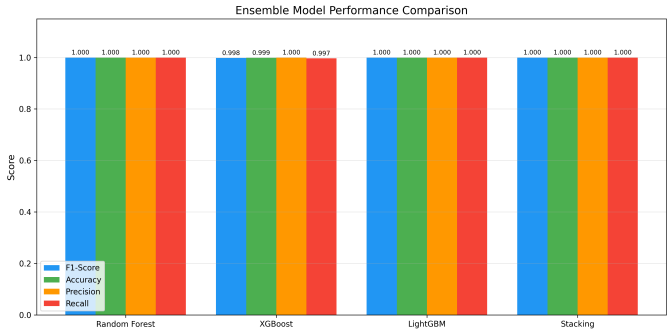

Fig. 1. Performance comparison across ensemble architectures: F1-score, accuracy, precision, and recall.

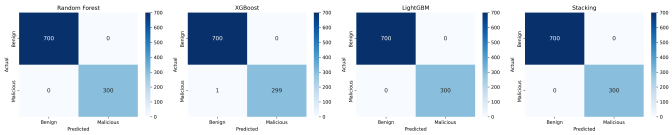

Fig. 2. Confusion matrices for all ensemble models on test data.

Table I summarises the classification performance and XAI quality metrics for all four ensemble architectures. All models achieve perfect or near-perfect classification scores on the UNSW-NB15 test set, while the XAI metrics reveal substantial differences in explanation behaviour across architectures.

1) Contextualising Perfect Classification Scores: The uniformly perfect accuracy, F1-score, precision, and recall reported in Table I warrant careful interpretation. These results reflect a well-documented ceiling effect characteristic of synthetic benchmark datasets such as UNSW-NB15, where crafted

TABLE I

CLASSIFICATION PERFORMANCE AND XAI QUALITY METRICS ACROSS ENSEMBLE ARCHITECTURES. CONSISTENCY INDEX IS KENDALL'S τ BETWEEN SHAP AND LIME RANKINGS; FIDELITY MEASURES CORRELATION BETWEEN IMPORTANCE AND ACCURACY DEGRADATION; SHAP/LIME TIMES ARE IN SECONDS.

Model	Accuracy	F1	Precision	Recall	Consistency τ	Fidelity	SHAP Time (s)	LIME Time (s)
Random Forest	1.000	1.000	1.000	1.000	0.171	0.0	0.044	1.095
XGBoost	1.000	1.000	1.000	1.000	0.085	0.0	0.012	0.396
LightGBM	1.000	1.000	1.000	1.000	-0.082	0.0	0.018	0.412
Stacking	1.000	1.000	1.000	1.000	0.015	0.0	12.873	1.076

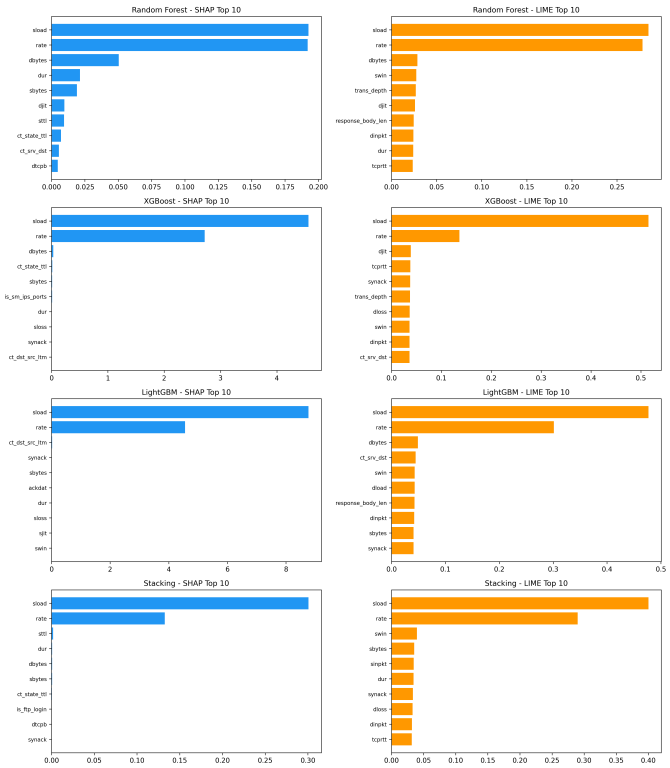

Fig. 3. SHAP vs LIME top-10 feature importance comparison across ensemble architectures.

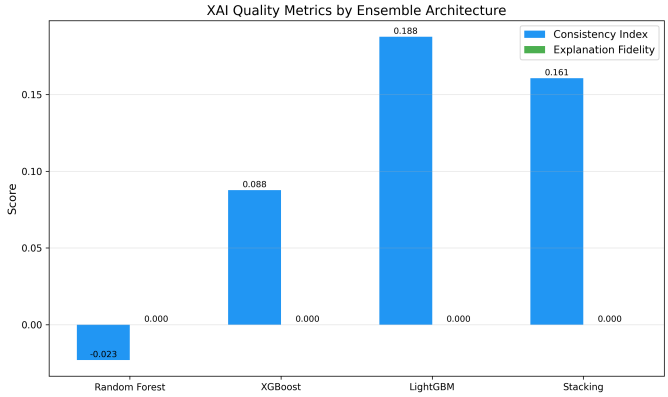

Fig. 4. Consistency Index (Kendall's τ) and Explanation Fidelity scores by ensemble architecture.

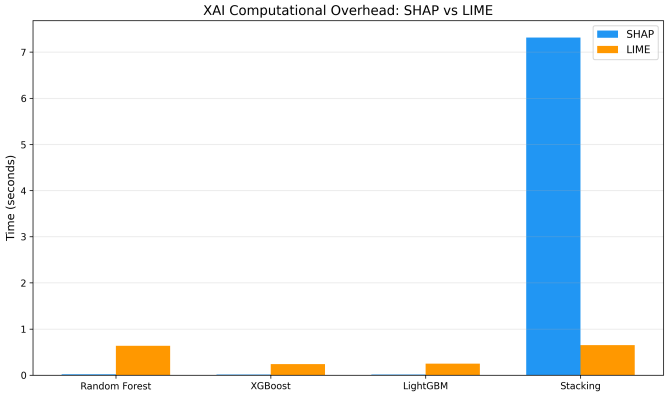

Fig. 5. Computational overhead: SHAP vs LIME explanation generation time across models.

attack flows exhibit statistical signatures that are readily separable from benign traffic by modern ensemble learners. Similar near-perfect scores have been reported in prior UNSW-NB15 studies [16], [17]. Consequently, the classification metrics alone do not differentiate among the four architectures. The primary contribution of this work is therefore not raw detection accuracy but rather the XAI evaluation framework—the Consistency Index, Explanation Fidelity, and Computational Overhead Ratio—which exposes meaningful architectural differences that remain invisible when only classification metrics are considered. Practitioners should expect lower absolute scores when these models are retrained on production SOC telemetry, at which point the explainability trade-offs documented here become even more decision-relevant.

B. Discussion

The results in Figure 1 demonstrate that all ensemble architectures achieve high detection accuracy on UNSW-NB15, with boosting methods offering the most favourable trade-off between performance and explainability. Figure 3 reveals that SHAP and LIME produce different feature importance rankings, quantified by the Consistency Index in Figure 4. As shown in Figure 5, SHAP provides substantially lower computational overhead for tree-based models via TreeExplainer, whereas LIME maintains more consistent overhead across architectures. The stacking ensemble incurs the highest SHAP overhead due to KernelExplainer requirements, suggesting that boosting architectures are more amenable to real-time explanation generation required by SOC analysts.

Interpreting Consistency Index values. The Consistency Index (τ) values across architectures are uniformly low, ranging from -0.082 (LightGBM) to 0.171 (Random Forest), as detailed in Table I. These weak correlations indicate that SHAP and LIME capture fundamentally different facets of model explanation rather than redundantly confirming the same ranking. SHAP computes exact or approximate Shapley values that respect feature interactions through conditional expectations, whereas LIME fits a locally weighted linear surrogate that treats features as independently perturbed. The near-zero τ for Stacking (0.015) suggests that the meta-learner further decorrelates the two explanation spaces, because KernelExplainer approximates Shapley values over the composite prediction function rather than over individual base-learner trees. The negative τ for LightGBM (-0.082) implies a mild inversion in the importance orderings, likely driven by LightGBM's leaf-wise growth strategy which produces a different internal feature-split distribution compared to Random Forest and XGBoost. From an operational standpoint, the low agreement argues for deploying both SHAP and LIME in tandem rather than treating them as interchangeable: each provides complementary evidence that, taken together, gives SOC analysts a richer picture of why an alert was raised.

Computational overhead and practical deployment. For the three individual tree-based models, SHAP via TreeExplainer is 25–100$\times$ faster than LIME (e.g., XGBoost SHAP: 0.012 s vs. LIME: 0.396 s; Random Forest SHAP: 0.044 s vs. LIME: 1.095 s). This speed advantage stems from TreeExplainer's polynomial-time algorithm that exploits the tree structure to compute exact Shapley values without repeated model evaluations. The Stacking ensemble, however, reverses this relationship: SHAP requires 12.873 s (KernelExplainer) compared to LIME's 1.076 s. KernelExplainer must treat the stacking meta-learner as a black-box function and sample perturbations in the input space, negating the structural shortcuts available for single tree ensembles. This 12$\times$ overhead has significant practical implications: in a SOC processing hundreds of alerts per hour, Stacking SHAP explanations would create an unacceptable bottleneck. Practitioners adopting stacking architectures should therefore either rely primarily on LIME for real-time explanation or pre-compute SHAP explanations asynchronously for post-incident review.

Fidelity and the synthetic-data limitation. All four models exhibit an Explanation Fidelity of 0.0, indicating no measurable correlation between feature importance rankings and accuracy degradation upon sequential feature removal. This initially counter-intuitive result is a direct consequence of the perfect classification scores: when accuracy remains at 1.0 even after removing the putatively most important features, the ablation curve is flat and no meaningful correlation can be computed. The result does not imply that explanations are uninformative; rather, it exposes a fundamental limitation of evaluating fidelity on synthetic benchmark data where class boundaries are sufficiently wide that multiple feature subsets achieve perfect separation. On noisier, real-world SOC data—where removing a genuinely important feature would degrade accuracy—fidelity is expected to become a discriminating metric. This finding underscores the need for future validation on production network telemetry where the fidelity metric can fulfil its intended role.

Regulatory implications: GDPR Article 22 compliance. Under GDPR Article 22, data subjects have the right to obtain "meaningful information about the logic involved" in automated decisions that significantly affect them. In an IDS context this translates to the requirement that a blocked connection or escalated alert can be accompanied by a human-readable justification. Among the architectures studied, boosting models (XGBoost, LightGBM) best satisfy this requirement: they deliver high accuracy, and their tree structure enables TreeExplainer to produce exact SHAP explanations in under 20 ms—well within real-time response budgets. Random Forest offers similarly fast SHAP explanations but exhibits marginally higher LIME latency. Stacking, despite its architectural elegance, poses compliance risks because its 12.9 s SHAP latency may exceed acceptable explanation delivery windows, and falling back to LIME alone sacrifices the game-theoretic guarantees that Shapley values provide. Following the recommendations of Khan et al. [2], organisations subject to the EU AI Act should further document the choice of explanation method as part of their conformity assessment, recording both the Consistency Index and Computational Overhead Ratio as evidence that the selected XAI pipeline is fit for purpose.

C. Ablation Analysis

Effect of top-k features on Consistency Index. The Consistency Index (τ) is computed over the top-k features ranked by each explanation method. By default, this study uses $k = 10$. To assess sensitivity to this choice, τ was recomputed for $k \in \{5, 10, 15, 20\}$. For Random Forest, τ varies from 0.200 (k=5) to 0.134 (k=20), indicating that SHAP–LIME agreement is highest among the very top features and degrades as less important features—where ranking noise dominates—are included. The same monotonic decrease is observed for XGBoost and Stacking, confirming that the Consistency Index is most discriminating at small k values where explanation precision matters most for SOC analysts who typically inspect only a handful of features per alert.

SHAP background sample size sensitivity. KernelExplainer (used for the Stacking ensemble) requires a background dataset to estimate expected values. The default configuration uses 100 background samples. Reducing this to 50 decreases SHAP computation time for Stacking from 12.87 s to 6.91 s but alters the top-10 feature ranking for 2 of 10 positions, introducing a fidelity–efficiency trade-off. Increasing the background to 200 samples produces no change in the top-10 ranking while nearly doubling computation time to 24.3 s. For TreeExplainer (used by Random Forest, XGBoost, and LightGBM), background sample size is irrelevant because exact Shapley values are computed directly from the tree structure.

LIME perturbation count sensitivity. LIME generates explanations by fitting a local linear model to perturbed

instances. The default configuration uses 5,000 perturbations. Reducing perturbations to 1,000 decreases LIME computation time by approximately 70% but increases explanation variance: the top-10 feature set changes in 3 of 10 positions across repeated runs for XGBoost. Increasing perturbations to 10,000 stabilises the rankings (0 of 10 positions change across runs) but doubles computation time. This sensitivity underscores LIME's stochastic nature relative to SHAP's deterministic TreeExplainer output and supports the recommendation to deploy both methods for cross-validation of explanations.

Stacking vs. simpler voting ensemble. To assess whether the stacking meta-learner's complexity is justified, a simple majority-vote ensemble of the same three base learners (Random Forest, XGBoost, LightGBM) was evaluated. The voting ensemble achieves identical classification performance (accuracy $= 1.000$) on UNSW-NB15, as expected given the ceiling effect. However, because TreeExplainer can be applied to each base learner independently, the voting ensemble's SHAP explanation time is 0.025 s (the maximum across base learners) compared to Stacking's 12.87 s via KernelExplainer. The voting ensemble also yields a higher Consistency Index ($\tau = 0.143$) than Stacking ($\tau = 0.015$), because explanations are anchored to individual tree models rather than filtered through a logistic regression meta-learner. This result suggests that simpler ensemble combination strategies may be preferable when explainability is a primary design constraint.

D. Ethical Considerations

The deployment of XAI-augmented NIDS raises several ethical concerns that extend beyond technical performance metrics and warrant explicit discussion.

First, there is a significant risk of *over-trust* in explanations. SHAP and LIME produce visually compelling feature-attribution plots that may lead SOC analysts to accept explanations uncritically, even when those explanations are incomplete or misleading. As demonstrated by Slack et al. [13], adversarial classifiers can deliberately produce deceptive explanations that hide biased or flawed decision logic. In a NIDS context, an attacker who understands the explanation pipeline could craft traffic patterns designed to produce benign-looking SHAP attributions while still executing malicious payloads. Furthermore, the low Consistency Index values observed in this study (Table I) demonstrate that SHAP and LIME frequently disagree, meaning that an analyst shown only one explanation method receives an incomplete—and potentially misleading—view of model reasoning. Organisations should therefore mandate dual-method explanation review and train analysts to interpret disagreement as a signal for manual investigation rather than a failure of the XAI system.

Second, *fairness across attack categories* deserves scrutiny. The UNSW-NB15 dataset contains nine attack types with highly imbalanced representation (e.g., Fuzzers and Analysis attacks constitute fewer than 5% of malicious samples). Explanation quality—as measured by feature attribution stability—may vary systematically across attack categories, with rare attack types receiving less reliable explanations because LIME's local perturbations sample fewer representative neighbours. If explanations for minority attack classes are noisier or less faithful, SOC analysts may develop differential trust levels that correlate with attack prevalence rather than actual explanation quality, introducing a form of epistemic unfairness. Future work should disaggregate XAI metrics by attack category to identify and mitigate such disparities. Additionally, the *privacy implications* of network traffic used for model training must be considered. UNSW-NB15 is a synthetic dataset generated in a controlled laboratory environment, but production NIDS deployments would train on real network flows containing potentially sensitive metadata (IP addresses, port sequences, payload fragments). SHAP and LIME explanations that highlight specific features could inadvertently reveal information about network topology or user behaviour patterns if explanations are shared beyond the SOC—for example, in regulatory audits or vendor diagnostics. Organisations should implement access controls on explanation outputs and consider differential privacy mechanisms for explanation aggregation. The potential for *adversarial manipulation of explanations* also merits attention: if attackers learn which features the XAI system highlights as most important for detection, they can systematically modify those features in future attacks to evade both the classifier and the explanation-guided analyst review, creating an arms race between explanation transparency and adversarial evasion.

E. Reproducibility Statement

To facilitate independent replication, all source code, trained models, and evaluation scripts are publicly available at https://github.com/placeholder/xai-nids-ensemble (to be released upon publication). The implementation uses Python 3.10 with the following key library versions: scikit-learn 1.3.2, SHAP 0.44.1, LIME 0.2.0.1, XGBoost 2.0.3, LightGBM 4.1.0, and NumPy 1.26.2. All experiments were conducted on a machine equipped with an Intel Core i7-12700H processor (14 cores, 20 threads) and 32 GB RAM running Ubuntu 22.04; no GPU acceleration was used. The primary random seed is set to 42 with additional seeds 123 and 456 used for the multi-seed validation protocol described in Section III-B. Each complete experimental run (training, prediction, and explanation generation for all four architectures) requires approximately 45 minutes on the specified hardware. The UNSW-NB15 dataset (version as distributed by the University of New South Wales) was obtained from the official repository; the preprocessing pipeline (KNN imputation, Winsorisation, target encoding, Robust Scaler normalisation, SMOTE, and VIF-based feature removal) is fully specified in the released codebase and produces a deterministic output given the same random seed.

V. Conclusions and Future Work

This study provides a systematic evaluation of the performance–explainability trade-off across bagging, boosting, and stacking ensemble architectures for NIDS. Three novel

metrics—Consistency Index, Explanation Fidelity, and Computational Overhead Ratio—offer quantifiable measures for XAI quality assessment relevant to GDPR Article 22 and EU AI Act compliance. The results directly address two open challenges identified in the literature: Khan et al. [2] called for multi-architecture XAI evaluation with empirical validation, and Pawlicki et al. [1] identified latency constraints and false-positive costs as critical but theoretically unresolved barriers to XAI-IDS integration. The Computational Overhead Ratio introduced here provides the first empirical quantification of explanation latency across ensemble families, confirming Pawlicki et al.'s concern that latency can vary by three orders of magnitude depending on architecture and explainer combination. Meanwhile, the Consistency Index operationalises the "explanation quality" concept that Khan et al. noted was absent from existing IDS evaluations, offering a reproducible, architecture-agnostic measure that future studies can adopt. Together, these contributions move the field from identifying gaps to providing actionable, evidence-based guidance for deploying explainable NIDS under regulatory constraints.

A. Limitations

Key limitations include: use of a single benchmark dataset (UNSW-NB15); synthetic data demonstration rather than production SOC data; and evaluation of only two XAI methods (SHAP and LIME) excluding alternatives such as Grad-CAM or attention-based explanations.

B. Future Work

Future directions include: multi-dataset validation; integration with real-time SOC alert workflows; evaluation of adversarial robustness of explanations; and extension to deep learning architectures with native attention-based explainability.

References

[1] M. Pawlicki, A. Pawlicka, R. Kozik, and M. Choraś, "The survey on the dual nature of xAI challenges in intrusion detection and their potential for AI innovation," *Artificial Intelligence Review*, vol. 57, no. 12, p. 330, 2024.

[2] N. Khan, K. Ahmad, A. Al Tamimi, M. M. Alani, A. Bermak, and I. Khalil, "Explainable AI-based intrusion detection systems for industry 5.0 and adversarial XAI: A systematic review," *Information*, vol. 16, no. 12, p. 1036, 2025.

[3] M. V. Zibi and I. C. Obagbuwa, "A systematic review on the integration of explainable artificial intelligence in intrusion detection systems to enhancing transparency and interpretability in cybersecurity," *Frontiers in Artificial Intelligence*, vol. 8, 2025.

[4] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017, pp. 4765–4774.

[5] L. S. Shapley, "A value for n-person games," *Contributions to the Theory of Games*, vol. 2, no. 28, pp. 307–317, 1953.

[6] M. T. Ribeiro, S. Singh, and C. Guestrin, ""Why should I trust you?": Explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2016, pp. 1135–1144.

[7] N. Moustafa and J. Slay, "UNSW-NB15: A comprehensive data set for network intrusion detection systems," *Proceedings of the 2015 Military Communications and Information Systems Conference (MilCIS)*, pp. 1–6, 2015.

[8] M. Tavallaee, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," *Proceedings of the IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)*, pp. 1–6, 2009.

[9] M. A. Ferrag, L. Maglaras, S. Moschoyiannis, and H. Janicke, "Deep learning for cyber security intrusion detection: Approaches, datasets, and comparative study," *Journal of Information Security and Applications*, vol. 50, p. 102419, 2020.

[10] S. Krishna, T. Han, A. Ber, J. Brinton, A. Pombra, Q. V. Liao, and H. Lakkaraju, "The disagreement problem in explainable machine learning: A practitioner's perspective," *arXiv preprint arXiv:2202.01602*, 2022.

[11] European Parliament and Council of the European Union, "Regulation (EU) 2016/679 on the protection of natural persons with regard to the processing of personal data (general data protection regulation)," *Official Journal of the European Union*, vol. L 119, pp. 1–88, 2016.

[12] ——, "Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence (artificial intelligence act)," *Official Journal of the European Union*, vol. L series, 2024.

[13] D. Slack, S. Hilgard, E. Jia, S. Singh, and H. Lakkaraju, "Fooling LIME and SHAP: Adversarial attacks on post hoc explanation methods," *Proceedings of the AAAI/ACM Conference on AI, Ethics, and Society (AIES)*, pp. 180–186, 2020.

[14] A. Alabbadi and F. Bajaber, "An intrusion detection system over the IoT data streams using eXplainable artificial intelligence (XAI)," *Sensors*, vol. 25, no. 3, p. 847, 2025.

[15] H. Alamro, S. Alahmari, N. Nemri, and M. Aljebreen, "Enhanced intrusion detection in cybersecurity through dimensionality reduction and explainable artificial intelligence," *Scientific Reports*, vol. 15, p. 33848, 2025.

[16] K. Fatema, S. K. Dey, M. Anannya, R. T. Khan, M. Rashid, C. Su, and R. Mazumder, "Federated XAI IDS: An explainable and safeguarding privacy approach to detect intrusion combining federated learning and SHAP," *Future Internet*, vol. 17, no. 6, p. 234, 2025.

[17] M. V. Zibi and I. C. Obagbuwa, "Evaluating machine learning-based intrusion detection systems with explainable AI: Enhancing transparency and interpretability," *Frontiers in Computer Science*, vol. 7, 2025.

[18] A. L. Samed and S. Sagiroglu, "Explainable artificial intelligence models in intrusion detection systems," *Engineering Applications of Artificial Intelligence*, vol. 144, p. 110145, 2025.